

1. Administrative & Study Identification

Full Study Title: A Study of Healthy Donor CD19-targeted Allogeneic CAR T Cells in Participants With Severe, Refractory Autoimmune Diseases **Short Title/Acronym:** IM060 **Protocol Number:** IM060-0001 **NCT Number:** NCT07115745 **Study Status:** Recruiting **Sponsor Name:** Juno Therapeutics, Inc., a Bristol-Myers Squibb Company **Investigational Product:** BMS-986515 (Allogeneic CD19-targeted CAR T cells) **Phase of Development:** Phase 1 **Medical Monitor:** BMS Clinical Trials Contact Center

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To determine the safety, tolerability, optimal dose, and preliminary efficacy of BMS-986515, a healthy donor (HD) allogeneic CAR T cell therapy. **Secondary Objectives:** To evaluate the clinical disease improvement across multiple severe autoimmune populations (SLE, IIM, SSc, RA), pharmacokinetics, and durability of response.

2.2 Background & Rationale To address the critical unmet medical need in patients with severe, refractory autoimmune diseases who have inadequate response or intolerance to steroids and standard immunosuppressive therapies. Utilizing BMS-986515, an allogeneic ("off-the-shelf") CD19-targeted CAR-T cell therapy, offers a potentially scalable, readily available, and cost-effective alternative to autologous CAR-T treatments, aiming to achieve a deep "immune reset" and long-term disease remission.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Single-arm (No Allocation) **Blinding:** Open-label **Randomization:** N/A **Trial Status:** Recruiting

3.2 Planned Enrollment & Duration Target **NS:** 125 participants **Number of Sites:** International (Multiple BMS Clinical Trial locations) **Estimated Study Start:** 04-Sep-2025 **Estimated Primary Completion:** 13-Feb-2029

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 18 years. 2. Confirmed diagnosis of severe, refractory autoimmune disease: Systemic Lupus Erythematosus (SLE) based on 2019 ACR/EULAR criteria (must

be positive for at least one of: anti-nuclear antibody, anti-dsDNA, anti-histone, anti-chromatin or anti-Sm antibody), Idiopathic Inflammatory Myopathies (IIM) meeting 2017 ACR/EULAR criteria, Systemic Sclerosis (SSc) meeting 2013 ACR/EULAR criteria, or difficult-to-treat Rheumatoid Arthritis (RA) meeting 2010 ACR/EULAR criteria. 3. Inadequate response or intolerance to standard of care therapies, including steroids and immunosuppressive treatments, with evidence of active disease at screening.

4.2 Exclusion Criteria 1. Pregnant or nursing women. 2. Active hepatitis B, C or HIV, or a prior history of malignancies. 3. Uncontrolled or active infection, history of certain cardiovascular conditions within 6 months, significant lung impairment, inadequate organ function, active central nervous system (CNS) disorders, or previous CAR-T cell therapy. 4. Any other systemic autoimmune disease (e.g., SLE caused by drugs, or combination of other autoimmune diseases).

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Single infusion of BMS-986515 (healthy donor CD19-targeted Allogeneic CAR T cells). **Route of Administration:** Intravenous (IV) infusion. **Duration of Treatment:** Single administration with extensive long-term safety and durability post-infusion follow-up.

5.2 Concomitant & Prohibited Therapy Permitted: Lymphodepleting conditioning regimen (Fludarabine and Cyclophosphamide [Trade names: Cytoxan, Endoxana]) prior to infusion; supportive medications like Tocilizumab for the management of Cytokine Release Syndrome (CRS). **Prohibited:** Previous CAR-T therapy; concurrent use of overlapping systemic immunosuppressants that could interfere with CAR-T cell expansion.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to -7	Informed Consent, Disease Activity Scoring, Safety Labs, Autoantibody Testing
Baseline	Day -6 to 0	Lymphodepletion therapy, Study Drug Infusion (BMS-986515)
Interim	Weeks 1-4	Acute safety monitoring (CRS/ICANS assessment), Safety Labs, Efficacy check
End of Study	Year 1 to 15	Long-term follow-up for gene/cellular therapy safety and Durability of Response

7. Outcome Measures (Endpoints)

7.1 Primary Endpoints Definition: 1. Number of participants with treatment-emergent adverse events (TEAEs). 2. Number of

participants with serious AEs (SAEs). 3. Number of participants with AEs of special interest (AESIs). **Measurement Method:** Clinical observation, standard laboratory evaluations, and established toxicity grading scales (e.g., ASTCT for CRS and neurotoxicity) to establish safety, tolerability, and optimal dose.

7.2 Secondary & Exploratory Endpoints Pharmacokinetics: 1. Maximum observed concentration (C_{max}). 2. Area under the concentration-time curve (AUC). 3. Time of maximum observed concentration (T_{max}). **Efficacy:** Preliminary clinical efficacy measured by improvements in disease-specific activity indices (e.g., ACR responses for RA, SLEDAI for SLE).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous inpatient/outpatient monitoring with rigorous specific screening for Cytokine Release Syndrome (CRS) and Immune Effector Cell-Associated Neurotoxicity Syndrome (ICANS). **Serious Adverse Events (SAEs):** Standard 24-hour reporting timelines, including extended long-term tracking for potential secondary malignancies per gene-editing cellular therapy guidelines. **Data Monitoring Committee (DMC):** Independent Data Safety Monitoring Board utilized for dose escalation decisions and toxicity review.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety Set (all participants who receive at least one dose of the lymphodepleting regimen or BMS-986515) and Efficacy Evaluable Population. **Sample Size Justification:** Target enrollment of $N=125$ is adequately sized for a Phase 1 study to observe early safety signals, determine the Recommended Phase 2 Dose (RP2D), and explore initial efficacy across the four distinct autoimmune cohorts. **Handling of Missing Data:** Standard handling mechanisms for early-phase oncology/immunology studies; patients withdrawing prior to initial DLT observation periods may be replaced.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Rigorous onsite source data verification and remote monitoring tailored for high-risk, advanced therapy medicinal products (ATMPs). **Audit Trail:** Strict documentation, eCRF locking, and rigorous chain of custody/identity tracking for the allogeneic cellular product.

11. Study Identifiers & Governance

Primary ID: IM060-0001 **Posting ID:** 1074692188349820121 **Registry Number:** NCT07115745 **Sponsor/Lead Organization:** Bristol Myers Squibb (Juno Therapeutics, Inc., a Bristol-Myers Squibb Company) **Study Title:** A Study of Healthy Donor CD19-targeted Allogeneic CAR T Cells in Participants With Severe, Refractory Autoimmune Diseases **Official Regulatory Title:** A Phase 1, Multicenter, Open-label Study of BMS-986515, Healthy Donor Allogeneic CD19-targeted Chimeric Antigen Receptor (CAR) T Cells, in Participants With Severe, Refractory Autoimmune Diseases **Preferred UMLS Name:** CD19-Directed Allogeneic Chimeric Antigen Receptor T-Cell Therapy

12. Clinical Framework (Autoimmune Specific)

Primary Condition: Severe, Refractory Autoimmune Diseases (SLE, SSc, IIM, RA) **Diagnosis Threshold:** Active disease presentation with specific clinical/laboratory criteria (e.g., autoantibodies for SLE) despite prior treatments **Medical Methodology:** B-cell depletion via CD19-directed Allogeneic CAR-T therapy **Optimization Target:** Complete immune reset leading to long-term, drug-free clinical remission

13. Study Procedures & Methodology

Management Pathway: Allogeneic cell therapy clinical pathway (screening, conditioning, infusion, and acute monitoring) **Education Protocol (HCP):** Site training for ATMP handling, CRS/ICANS grading, and rapid tocilizumab/steroid deployment **Education Protocol (Patient):** Comprehensive counseling on cellular therapy risks (infections, neurological events) and necessary post-infusion geographic proximity to the clinical site **Quality Audit Mechanism:** Continuous safety signal evaluation and cellular product chain-of-custody tracking

14. Observation & Follow-Up Schedule

Total Duration: Primary completion approximately 3.5 years (primary endpoint), with mandated 15-year long-term follow-up (LTFU) for gene-edited cellular therapies. **Onsite Visit Cadence:** Frequent evaluations during the acute phase (Days 1–28 post-infusion), transitioning to quarterly, and subsequently annual visits. **Remote Monitoring Frequency:** Intermittent telehealth and centralized lab tracking after the initial acute safety window. **Checklist Review Interval:** Continuous AE tracking and disease symptom diaries updated per protocol intervals.

15. Sign-off & Approval

Role	Name	Signature	Date	:	---	:	---	:	---	:	---
Principal Investigator	[Name]	_____	03-Apr-2026	:		:		:		:	
Clinical Operations Lead	[Name]	_____	03-Apr-2026	:		:		:		:	Lead
Statistician	[Name]	_____	03-Apr-2026	:		:		:		:	